

Task-4

Developing queries with OML Multi-row Functions and operators

perform the advanced query processing (multi row abilities) and test its heuristics using the designing of (Optimal correlated and nested subqueries such as finding summary statistics).

Consider the schema for

EMPLOYEES (emp-no, emp-Name, department, deptno, salary, age)

ORDERS (emp-no, order id, price, qty-ord, qty-hand)

ITEMS (itemid, itemname, qty-ord, qty-hand, itemrate)

Queries using OML, INTERSECT, MINUS.

Union: The union operator returning all distinct rows selected by two or more queries.

SQL> select emp-no from employees;

SQL> select emp-no from orders;

SQL> select emp-no from employees union select emp-no from orders;

Union All:

SQL> select emp-no from employees union all select emp-no from orders;

Intersect:

SQL> select emp-no from employees intersect select emp-no from orders;

Minus:

SQL> select emp-no from employees minus select emp-no from orders;

Practice questions:

1. Find the emp-no of employees whose name starts with 'S' and ends with 'H'.
2. Find the names of the employees whose age is between 20 and 30.
3. Display all the name of the employees beginning with 'V'.
4. Display the sorted list of employees names.

Queries using Group By, Having clause and Order clause.

GROUP BY: This query is used to group to all the records in a relation to gether for each and every value of a specific key(s) and then display them for a selected set of fields in the relation.

SQL > Select deptno, count(*) from employees group by deptno;

GROUP BY-HAVING: The HAVING clause was added to SQL because the WHERE keyword could not be used with aggregate functions.

The HAVING clause must follow the GROUP BY clause in a query and must also precede the ORDER BY clause if used.

SQL > Select deptno, count(*) from employees group by deptno having count(*) > 10;

ORDER BY: This query is used to display a selected set of fields from a relation in an ordered manner based on some field.

Syntax:

Select (column(s) from Table Name where [condition(s)] [Order by column(s)]

SQL > select empno, ename, salary from employees order by salary;
SQL > select empno, emp-ename, salary from employees order by salary desc;

SQL > * plus having following operators

SQL > select salary + comm from emp-master;
emp + comm

SQL > select salary + comm net-sal from emp-master;

SQL > select 12 * (salary + comm) annual-net-sal from emp-master;

Subqueries:

SQL > select * from employees

SQL > insert into employees select * from employees where emp-id in (select emp-id from employees)

SQL > Update employees set salary = salary * 10 where department in (select department from employees where department = 'sales');
Delete from employees where department in (select department from employees where department = 'sales');

IN:

Query: SELECT * FROM employees WHERE department in ('sales', 'marketing');

NOT IN:

Query: SELECT * FROM employees WHERE department NOT in ('sales', 'marketing');

Exists:

Query: SELECT * FROM employees WHERE EXISTS (SELECT * FROM orders WHERE emp-no = (link unavailable));

NOT EXISTS:

Query: SELECT * FROM employees WHERE NOT EXISTS (SELECT * FROM orders WHERE orders.emp-no = (emp.unavailable));

ALL:

Query: SELECT * FROM employees WHERE salary > ALL (SELECT salary FROM employees WHERE department = 'sales');

ANY:

Query: SELECT * FROM employees WHERE salary > ANY (SELECT salary FROM employees WHERE department = 'sales');

Sales Select * from Order-master where order-no = (select order-no from orders where order-no = '0001');

Sales select * from order-master where order-no = (select order-no from orders);

Sales select * from order-master where order-no = any(select order-no from order-detail);

Sales select * from order-master where order-no in (select order-no from order-detail);

Sales select * from order-detail where qty-ord = all(select qty-hand from itemfile where itemrate > 250);

VEL TECH	
EX NO.	4
PERFORMANCE (5)	5
RESULT AND ANALYSIS (5)	5
VIVA VOCE (5)	5
RECORD (5)	5
TOTAL (20)	20
DATE	

Palata

RESULT: Thus, the developing queries with DML functions and operators

SQL*Plus: Release 11.2.0.2.0 Production on Thu Sep 11 13:59:48 2025

Copyright (c) 1982, 2014, Oracle. All rights reserved.

SQL> connect

Enter user-name: system

Enter password:

Connected.

SQL> create table employees(emp_no number(4), emp_name varchar(15), department
varchar(5), dept_no number(5), salary number(5), age number(2));

Table created.

SQL> desc employees

Name	Null?	Type
EMP_NO		NUMBER(4)
EMP_NAME		VARCHAR2(15)
DEPARTMENT		VARCHAR2(5)
DEPT_NO		NUMBER(5)
SALARY		NUMBER(5)
AGE		NUMBER(2)

SQL> create table orders(emp_no number(4), order_id number(5), price number(5), qty_ord
number(5), qty_hand number(5));

Table created.

SQL> desc orders

Name	Null?	Type
EMP_NO		NUMBER(4)
ORDER_ID		NUMBER(5)
PRICE		NUMBER(5)
QTY_ORD		NUMBER(5)
QTY_HAND		NUMBER(5)

SQL> create table itemfile(item_id number(5), item_name varchar(10), qty_ord
number(5), qty_hand number(5), item_rate number(5));

Table created.

SQL> desc itemfile

Name	Null?	Type
ITEM_ID		NUMBER(5)
ITEM_NAME		VARCHAR2(10)
QTY_ORD		NUMBER(5)
QTY_HAND		NUMBER(5)
ITEM_RATE		NUMBER(5)

SQL> insert into employees values(21,'koti','cseds',126,60000,21);

1 row created.

SQL> insert into employees values(22,'shivapriya','csed',128,80000,19);

1 row created.

SQL> insert into employees values(23,'maha','aids',142,70000,18);

1 row created.

SQL> insert into employees values(24,'bhasker','cseai',123,50000,20);

1 row created.

SQL> insert into employees values(25,'chandrajith','cseai',123,90000,21);

1 row created.

SQL> select * from employees;

EMP_NO	EMP_NAME	DEPAR	DEPT_NO	SALARY	AGE
21	koti	cseds	126	60000	21
22	shivapriya	csed	128	80000	19
23	maha	aids	142	70000	18
24	bhasker	cseai	123	50000	20
25	chandrajith	cseai	123	90000	21

SQL> insert into orders values(21,789,500,5,4);

1 row created.

SQL> insert into orders values(22,456,400,6,5);

1 row created.

SQL> insert into orders values(23,123,600,4,3);

1 row created.

SQL> insert into orders values(24,159,200,8,7);

1 row created.

SQL> insert into orders values(25,357,900,3,2);

1 row created.

SQL> select * from orders;

EMP_NO	ORDER_ID	PRICE	QTY_ORD	QTY_HAND
21	789	500	5	4
22	456	400	6	5
23	123	600	4	3
24	159	200	8	7
25	357	900	3	2

SQL> insert into itemfile values(1236,'dog',5,5,20000);

1 row created.

SQL> insert into itemfile values(14324,'gold',4,4,30000);

1 row created.

SQL> insert into itemfile values(5478,'pig',5,6,10000);

1 row created.

SQL> insert into itemfile values(9874,'bajji',10,9,1000);

1 row created.

SQL> insert into itemfile values(2854,'silve',6,5,60000);

1 row created.

SQL> select * from itemfile;

ITEM_ID	ITEM_NAME	QTY_ORD	QTY_HAND	ITEM_RATE
1236	dog	5	5	20000
14324	gold	4	4	30000
5478	pigy	5	6	10000
9874	baji	10	9	1000
2854	silve	6	5	60000

SQL> select 22 from employees;

22

22
22
22
22
22
22

SQL> select emp_no from employees
2 ;

EMP_NO

21
22
23
24
25

SQL> select emp_no from orders;

EMP_NO

21
22
23
24
25



SQL> select emp_no from employees union select emp_no from orders;

EMP_NO
21
22
23
24
25

SQL> select emp_no from employees union all select emp_no from orders;

EMP_NO
21
22
23
24
25
21
22
23
24
25

10 rows selected.

SQL> select emp_no from employees intersect select emp_no from orders;

EMP_NO
21
22
23
24
25

SQL> select emp_no from employees minus select emp_no from orders;

no rows selected

SQL> select dept_no, count(*) from employees group by dept_no;

DEPT_NO COUNT(*)

123	2
128	1
142	1
126	1

SQL> select dept_no,count(*) from employees group by dept_no having dept_no is not null;

DEPT_NO COUNT(*)

123	2
128	1
142	1
126	1

SQL> select emp_no,emp_name,age,salary from employees order by salary desc;

EMP_NO	EMP_NAME	AGE	SALARY
25	chandrajith	21	90000
22	shivapriya	19	80000
23	maha	18	70000
21	koti	21	60000
24	bhasker	20	50000

SQL> select salary+age from employees;

SALARY+AGE

60021
80019
70018
50020
90021

SQL> select * from employees;

EMP_NO	EMP_NAME	DEPAR	DEPT_NO	SALARY	AGE
21	koti	cse	126	60000	21
22	shivapriya	cse	128	80000	19
23	maha	ai	142	70000	18
24	bhasker	cse	123	50000	20

25 chandrajith cseai 123 90000

SQL> insert into employees select * from employees where emp_no in (select emp_no from employees);

5 rows created.

SQL> select * from employees;

EMP_NO	EMP_NAME	DEPAR	DEPT_NO	SALARY	AGE
21	koti	cseai	126	60000	21
22	shivapriya	cseai	128	80000	19
23	maha	ai	142	70000	18
24	bhasker	cseai	123	50000	20
25	chandrajith	cseai	123	90000	21
21	koti	cseai	126	60000	21
22	shivapriya	cseai	128	80000	19
23	maha	ai	142	70000	18
24	bhasker	cseai	123	50000	20
25	chandrajith	cseai	123	90000	21

10 rows selected.

SQL> update employees set salary=salary*10 where department in (select department from employees where department='sales');

0 rows updated.

SQL> update employees set salary=salary*1 where department in (select department from employees where department='cseai');

2 rows updated.

SQL> select * from employees;

EMP_NO	EMP_NAME	DEPAR	DEPT_NO	SALARY	AGE
21	koti	cseai	126	60000	21
22	shivapriya	cseai	128	80000	19
23	maha	ai	142	70000	18
24	bhasker	cseai	123	50000	20
25	chandrajith	cseai	123	90000	21

21	koti	cse	126	60000	21
22	shivapriya	ced	128	80000	19
23	maha	aids	142	70000	18
24	bhasker	cseai	123	50000	20
25	chandrajith	cseai	123	90000	21

10 rows selected.

SQL> select * from employees where department in ('cse','ced');

EMP_NO	EMP_NAME	DEPAR	DEPT_NO	SALARY	AGE

21	koti	cse	126	60000	21
22	shivapriya	ced	128	80000	19
21	koti	cse	126	60000	21
22	shivapriya	ced	128	80000	19

SQL> select * from employees where department not in ('cse','ced');

EMP_NO	EMP_NAME	DEPAR	DEPT_NO	SALARY	AGE

23	maha	aids	142	70000	18
24	bhasker	cseai	123	50000	20
25	chandrajith	cseai	123	90000	21
23	maha	aids	142	70000	18
24	bhasker	cseai	123	50000	20
25	chandrajith	cseai	123	90000	21

6 rows selected.

SQL> select * from employees where exists (select * from orders where orders.emp_no=22);

EMP_NO	EMP_NAME	DEPAR	DEPT_NO	SALARY	AGE

21	koti	cse	126	60000	21
22	shivapriya	ced	128	80000	19
23	maha	aids	142	70000	18
24	bhasker	cseai	123	50000	20
25	chandrajith	cseai	123	90000	21
21	koti	cse	126	60000	21
22	shivapriya	ced	128	80000	19
23	maha	aids	142	70000	18
24	bhasker	cseai	123	50000	20
25	chandrajith	cseai	123	90000	21

10 rows selected.

SQL> select * from employees where not exists (select * from orders where orders.emp_no=22);

no rows selected

SQL> select * from employees where not exists (select * from orders where orders.emp_no=24);

no rows selected

SQL> select * from employees where not exists (select * from orders where orders.emp_no=21);

no rows selected

SQL> select * from employees where not exists (select * from orders where orders.emp_no=22);

no rows selected

SQL> select * from employees where exists (select * from orders where orders.emp_no=24);

EMP_NO	EMP_NAME	DEPAR	DEPT_NO	SALARY	AGE
21	koti	cse	126	60000	21
22	shivapriya	cse	128	80000	19
23	maha	ai	142	70000	18
24	bhasker	cse	123	50000	20
25	chandrajith	cse	123	90000	21
21	koti	cse	126	60000	21
22	shivapriya	cse	128	80000	19
23	maha	ai	142	70000	18
24	bhasker	cse	123	50000	20
25	chandrajith	cse	123	90000	21

10 rows selected.

SQL> select * from employees where salary > all (select salary from employees where department='cse');

EMP_NO	EMP_NAME	DEPAR	DEPT_NO	SALARY	AGE
25	chandrajith	cseai	123	90000	21
25	chandrajith	cseai	123	90000	21

SQL> select * from orders where order_id=(select order_id from orders where order_id='123')

EMP_NO	ORDER_ID	PRICE	QTY_ORD	QTY_HAND
23	123	600	4	3

SQL> select * from orders where order_id=any(select order_id from orders);

EMP_NO	ORDER_ID	PRICE	QTY_ORD	QTY_HAND
21	789	500	5	4
22	456	400	6	5
23	123	600	4	3
24	159	200	8	7
25	357	900	3	2

SQL> select * from orders where order_id in(select order_id from orders);

EMP_NO	ORDER_ID	PRICE	QTY_ORD	QTY_HAND
21	789	500	5	4
22	456	400	6	5
23	123	600	4	3
24	159	200	8	7
25	357	900	3	2

SQL> select salary+price from employees,orders
2 ;

SALARY+PRICE
60500
80500
70500
50500
90500
60500
80500
70500
50500

90500
60400

SALARY+PRICE

80400
70400
50400
90400
60400
80400
70400
50400
90400
60600
80600

SALARY+PRICE

70600
50600
90600
60600
80600
70600
50600
90600
60200
80200
70200

SALARY+PRICE

50200
90200
60200
80200
70200
50200
90200
60900
80900
70900



50900

SALARY+PRICE

~~50900~~